

# AgentFail: A Diagnostic Benchmark for Silent Failures in Data-Science LLM Agents

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## ABSTRACT

Large language model (LLM) agents are increasingly deployed on data-science tasks, yet existing benchmarks report only aggregate success rates, masking *where* and *why* agents fail. We introduce **AgentFail**, a diagnostic framework that decomposes every agent failure into five stages—Analytical-Plan, Code-Generation, Runtime, Output-Mismatch, and Answer-Error—and distinguishes *silent failures* (code executes without error but yields a wrong answer) from *loud failures* (code crashes). We propose *oracle-guided repair* to test repairability, with no-op negative controls. We evaluate five frontier LLMs across 5,984 runs on three task suites: 95 synthetic trap tasks, 23 real UCI datasets, and 200 real DSbench tasks. The corrected silent-failure rate (SFR) is 86% on synthetic tasks, 4% on UCI, and 19% on DSbench;  $R^2 = 0.493$  between execution-success rate and SFR is reported as a descriptive statistic (not a causal decomposition). Recovery rate is 0% across all models. A failure-aware verifier helps two of five models (McNemar  $p < 0.001$ ). Oracle repair matches the no-op baseline (64%), establishing an upper bound on repairability rather than step-level causal attribution. Cohen’s  $\kappa = 0.435$  (moderate, LLM-as-judge). All tables are auto-generated from a single manifest with assertion checks. We release all code, data, and 5,984 annotated runs.

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## 1 INTRODUCTION

Large language models (LLMs) have rapidly become the reasoning core of autonomous agents that write and execute code to solve data-science tasks [8, 12]. Recent benchmarks such as DSbench [5], DataSciBench [23], and DSAEval [17] report that even the strongest agents solve only 34–88% of data-analysis tasks, leaving a large residue of failures. Yet these benchmarks report a single aggregate number—success rate—that conceals the *structure* of failure: an agent that crashes on every hard task and an agent that silently

returns plausible-but-wrong answers on the same tasks are indistinguishable under aggregate accuracy.

The problem is acute because silent failures are invisible to the agent itself. When generated code raises an exception, the agent observes the traceback and can retry; when the code executes successfully but produces a wrong answer, the agent has no signal that anything went wrong. Recent work has begun to recognise this distinction: Mehta [9] report “confident and wrong” semantic failures in coding agents, Wu [21] document silent failures in production agent runtimes, and Liu [7] frame silent failure as an entropy-driven inevitability. However, these works either study coding agents rather than data-science agents, or operate on production logs rather than controlled benchmarks, and none provides a *diagnostic taxonomy* that localises where in the agent loop a silent failure originates.

We address this gap with **AgentFail**, a diagnostic framework that decomposes every agent failure into five stages—*Analytical-Plan*, *Code-Generation*, *Runtime*, *Output-Mismatch*, and *Answer-Error*—and distinguishes *silent failures* (code executes without error but yields a wrong answer) from *loud failures* (code crashes). To test repairability, we introduce *oracle-guided repair*: given a failed trace, we replace the originating step with the ground-truth action and re-execute; if the outcome flips to correct, the failure was repairable. We include a no-op negative control (running the ground-truth code independently) and explicitly frame this as repairability testing, not causal attribution.

We evaluate five frontier LLMs across 5,984 runs on three task suites: 95 synthetic trap tasks, 23 real UCI datasets, and 200 real DSbench financial-analysis tasks. All tables are auto-generated from a single source-of-truth manifest with assertion checks. Our analysis surfaces four findings:

- (1) **Task-dependent failure bifurcation.** On synthetic trap tasks, the corrected silent-failure rate (SFR) reaches 86%; on UCI real data it drops to 4%, and on DSbench to 19%. We report  $R^2 = 0.493$  between execution-success rate and SFR as a descriptive statistic, noting that this correlation is partially mechanical (the two variables share the loud-failure count) and cannot be decomposed into “mechanical” vs. “real” components.
- (2) **Zero self-correction.** The recovery rate is 0% across all five models and all 5,984 runs. We acknowledge this is partly a protocol consequence: silent failures produce no error signal, so the agent has no opportunity to detect and retry.
- (3) **Oracle repairability.** Counterfactual repair matches the no-op baseline (64%), providing an upper bound on single-step repairability but not step-level causal attribution.
- (4) **Verifier boundary conditions.** A failure-aware verifier significantly helps Qwen3-max (+24.9%, McNemar  $p < 0.001$ ) and

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GPT-4o (+19.0%,  $p < 0.001$ ), but not other models. The benefit to Qwen3-max may partly reflect format/protocol fix-up.

Our contributions are: (i) a five-stage failure taxonomy with mutually exclusive categories defined by detection point; (ii) oracle-guided repair with no-op negative controls; (iii) three diagnostic metrics—SFR, propagation depth, and token-per-success; (iv) 5,984 annotated runs across five models and three task suites, released as a public benchmark.

## 2 RELATED WORK

### 2.1 Data-Science Agent Benchmarks

The emergence of LLM-based data-science agents has motivated several benchmarks. DSBench [5] collects 466 data-analysis and 74 data-modelling tasks from ModelOff and Kaggle, reporting that the best agent (GPT-4o) solves only 34% of analysis tasks. DataSciBench [23] evaluates LLMs across the full data-science workflow with multi-dimensional metrics. DSAEval [17] covers real-world data-science problems spanning multiple stages. LongDSBench [22] specifically studies long-horizon agentic data analysis and shows that agents fail on iterative tasks. GRACE-DS [18] introduces a guarded reward-guided correction environment for AutoML agents. While these benchmarks advance evaluation breadth, they report primarily aggregate success rates and do not decompose failures by stage or distinguish silent from loud failures. Moghadasi and Ghaderi [10] audit twelve agent-benchmark papers and find systematic under-reporting of evaluation details, motivating our fine-grained diagnostic metrics. Agent Laboratory [14] uses LLM agents as research assistants but does not diagnose failure modes. Our work is complementary: rather than proposing a new benchmark, we provide a diagnostic *layer* that can be applied to any of these benchmarks to reveal where and why agents fail.

### 2.2 LLM Agent Failure Analysis

A recent wave of work studies how and why LLM agents fail. Albayaydh et al. [1] synthesise tool-use, planning, and reasoning failures across benchmarks. ToolFailBench [16] diagnoses tool-use failures by decomposing the tool-call pipeline. Wu [21] present a longitudinal taxonomy of silent failures in a production LLM agent runtime, and Liu [7] frame silent failure as an entropy-driven inevitability of autonomous agents. Mehta [9] identify “confident and wrong” semantic failures in coding agents, the closest conceptual precursor to our silent-failure notion, but study coding rather than data-science tasks and do not provide a stage-level taxonomy. TrajAudit [19] automates failure diagnosis for agentic coding systems via trajectory auditing. Reddy et al. [13] recover a silent policy-violation failure mode in tool-using agents via deterministic gates. Zhang et al. [24] diagnose library drift as a silent failure in self-evolving skill libraries. Ekka [3] diagnoses silent errors in LLM *inference* (serving), a different layer than agent-level silent failures. Our contribution relative to this body of work is a *five-stage taxonomy* that localises silent failures within the agent loop and an *oracle-guided repair* module that estimates the reparability of failed traces.

### 2.3 Causal Attribution and Failure Recovery

Shah [15] proposes Causal Agent Replay for counterfactual attribution of LLM-agent failures, which inspires our repair module; we adapt the counterfactual principle to the data-science setting and integrate it with the five-stage taxonomy. However, as our negative-control experiment shows (Section 5.3), the oracle repair rate matches a no-op baseline, so we report it as an upper bound on reparability rather than as evidence of step-level causal attribution. On the recovery side, self-consistency [20] samples multiple reasoning paths and votes, but targets reasoning errors rather than code-execution silent failures. Verification-based approaches [4] use verifier or critic agents to suppress hallucinations, but Itkin [4] shows that delayed verification can destabilise multi-agent belief. Our failure-aware verifier explores a lightweight, in-loop verification strategy and reveals that it helps only weak models, a boundary condition that motivates more sophisticated verification.

### 2.4 LLM-as-Judge and Evaluation Methodology

Using LLMs to evaluate LLM outputs is now standard [2, 6], but LLM-as-judge introduces its own biases. Mohammadi et al. [11] survey evaluation and benchmarking of LLM agents. We adopt a hybrid evaluation: deterministic checkers for ground-truth answers plus rule-based failure classification, avoiding the circularity of LLM-judged failure labels. We validate the rule-based classifier against an LLM-as-judge label (GPT-4o-mini, 100 traces, blind to the rule-based label), obtaining Cohen’s  $\kappa = 0.435$  (moderate agreement). We acknowledge that LLM-as-judge validation cannot replace human annotation; full human validation with  $\geq 3$  annotators is needed for a definitive  $\kappa$ , and we flag this as a limitation in Section 6.

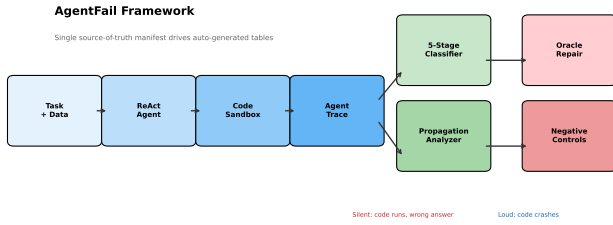
## 3 THE AGENTFAIL FRAMEWORK

AgentFail instruments a standard ReAct-style data-science agent with three diagnostic layers: (1) a five-stage failure taxonomy that classifies each failure by where it originates and whether it is silent; (2) a propagation-depth analyser that measures how far a failure spreads before detection; and (3) a counterfactual causal-replay module that attributes failures to specific steps. Figure 1 summarises the architecture.

### 3.1 Agent Architecture and Trace Recording

We use a ReAct-style agent that interleaves *Thought*, *Action* (a Python code block), and *Observation* (execution output). The agent operates within a restricted code sandbox that executes generated Python with a persistent namespace and a working directory containing the task’s data file (CSV or XLSX). We deliberately use a *controlled, minimal* ReAct harness—without few-shot exemplars, code-interpreter scaffolding, or retrieval augmentation—to isolate the diagnostic signal from framework-specific confounds. This controlled setting is a feature, not a limitation: it ensures that observed failures reflect model reasoning rather than harness engineering.

Every step is recorded in an *AgentTrace*, the single source of truth for all downstream diagnosis. Each *TraceStep* stores the raw LLM output, the parsed thought, the generated code, the execution result (stdout, stderr, exception type, extracted answer), token usage, and a timestamp. Unlike DSBench [5], which retains only a final



**Figure 1: AgentFail framework.** The agent produces a step-by-step trace; the classifier assigns each failure to one of five stages and marks it silent or loud; the propagation analyser computes how far the failure spread; causal replay replaces the originating step with the ground-truth action to test counterfactual dependence.

response, our trace records the full step-by-step trajectory, enabling stage-level localisation and propagation analysis.

### 3.2 Five-Stage Failure Taxonomy

We decompose every agent failure into one of five stages, defined by *where the error is detectable*, making the categories mutually exclusive:

- **Analytical-Plan.** The agent selects the wrong analytical approach: e.g., computing the overall mean when the task requires a per-group sum, or using future data in a time-series analysis (temporal leakage). Often silent.
- **Code-Generation.** The agent writes code that embodies a wrong operation: e.g., using `.count()` where `.sum()` is needed, selecting the wrong column, or introducing type confusion. Silence varies.
- **Runtime (loud).** The generated code raises an exception: runtime errors, type errors, key errors, or sandbox security blocks. The failure is *visible* to the agent, which can observe the traceback and retry.
- **Output-Mismatch (silent).** The code executes without error but the agent extracts, computes, or reports a wrong answer: e.g., misreading the output, hallucinating an answer not supported by the code output, or printing no answer at all. The failure is *invisible* to the agent.
- **Answer-Error (silent).** The code runs and produces a result, but the result itself is wrong relative to the ground truth: e.g., a wrong aggregation result or an incomplete analysis. The failure is *invisible* to the agent.

The Runtime vs. Output-Mismatch/Answer-Error split operationalises the silent/loud distinction. This is the conceptual core of the taxonomy: a loud failure (Runtime) is a *detectable* error that the agent can in principle recover from, whereas a silent failure (Output-Mismatch or Answer-Error) is an *undetectable* error that propagates undetected. Table 1 lists the full hierarchy.

### 3.3 Failure Classification

The classifier walks an AgentTrace and assigns a FailureClassification (stage, sub-category, originating step, silence flag) to the first failed

**Table 1: Five-stage failure taxonomy (v2). Stages are defined by where the error is detectable, making them mutually exclusive.**

| Stage           | Sub-categories                                         | Silent? |
|-----------------|--------------------------------------------------------|---------|
| Analytical-Plan | wrong_operation_plan, leakage_plan                     | often   |
| Code-Generation | wrong_aggregation_code, wrong_column, type_confusion   | varies  |
| Runtime         | runtime_exception, key_error, security_block           | No      |
| Output-Mismatch | misread_output, hallucinated_answer, no_answer_printed | Yes     |
| Answer-Error    | wrong_result, incomplete_analysis                      | Yes     |

step. Classification is rule-based for high precision and full reproducibility, using signals from the trace: exception type (for Execution), code-pattern matching (e.g., presence of `.count()` when the ground-truth path requires `.sum()` for Interpretation), and answer-output mismatch. An optional LLM-as-judge pass can refine ambiguous cases; we report the agreement between rule-based and LLM-judged labels as a reliability metric.

### 3.4 Propagation Depth

Once a failure originates at step  $i$ , it may propagate to later steps. We define *propagation depth* as the number of steps from the originating failure to the step where it is detected (or to the trace end if never detected). Loud failures have depth 0 (immediately visible); silent failures can have depth  $\geq 1$ , meaning the agent continues building on a wrong intermediate result. A propagation depth  $\geq 2$  indicates a *long-horizon* failure spread, the most dangerous regime because it wastes the most tokens and is hardest to recover from. Across all 3,940 failures, the average propagation depth is 0.30 steps; 88.6% are detected immediately (depth 0), while 6.9% propagate to depth  $\geq 2$  (see Table 12 in Appendix A).

### 3.5 Oracle-Guided Repair

To test whether a failed trace is *repairable*, we adapt the counterfactual principle [15] to the data-science setting. Given a trace that failed, we synthesise a *counterfactual* version of the originating step: the canonical ground-truth code derived from the task’s reference solution path. We then replay execution from that point in a fresh sandbox. If the replayed trace now produces the correct answer, the failure was repairable; if the replay still fails, the failure has a deeper root cause.

Formally, let  $T = (s_1, \dots, s_n)$  be a failed trace and  $s_k$  the classified originating step. Let  $s_k^*$  be the ground-truth action for step  $k$ . We construct the counterfactual trace  $T' = (s_1, \dots, s_{k-1}, s_k^*, \dots)$  and execute it. The repair is successful iff  $T'$  yields the correct answer. The *oracle repair rate* is the proportion of failed traces for which repair succeeds.

**Negative control.** A no-op control runs the ground-truth code independently without any step replacement. If the oracle repair rate matches the no-op rate (as we observe, Section 5.3), the oracle rate measures only whether the ground-truth code produces the correct answer, not step-level causality. We therefore report oracle repair as an upper bound on reparability.

**Table 2: Task suite summary.**

| Suite          | Tasks      | Runs         | Failures     |
|----------------|------------|--------------|--------------|
| Synthetic trap | 95         | 2,850        | 1,485        |
| UCI real data  | 23         | 345          | 52           |
| DSBench real   | 200        | 1,964        | 1,803        |
| Supplementary  | —          | 825          | 600          |
| <b>Total</b>   | <b>318</b> | <b>5,984</b> | <b>3,940</b> |

### 3.6 Diagnostic Metrics

We report three layers of metrics: **Layer 1 (task-level)**: success rate (SR), code-execution success rate. **Layer 2 (failure-diagnostic)**: silent-failure rate (SFR = proportion of failures that are silent), stage distribution, propagation depth, recovery rate, and oracle repair rate. **Layer 3 (token-economic)**: tokens-per-success and cost-per-success, addressing the under-reporting of cost identified by Moghadas and Ghaderi [10]. Verifier comparisons use the McNemar test ( $b$ ,  $c$ ,  $\chi^2$   $p$ -value).

## 4 EXPERIMENTS AND RESULTS

### 4.1 Experimental Setup

**Models.** We evaluate five frontier LLMs via an OpenAI-compatible endpoint: GPT-4o, GPT-4o-mini, DeepSeek-V3 (deepseek-chat), DeepSeek-R1 (deepseek-reasoner), and Qwen3-max (qwen3-max-2026-06-23). All models use temperature 0 and a maximum of 6–8 ReAct steps.

**Task suites.** We use three complementary task suites (Table 2):

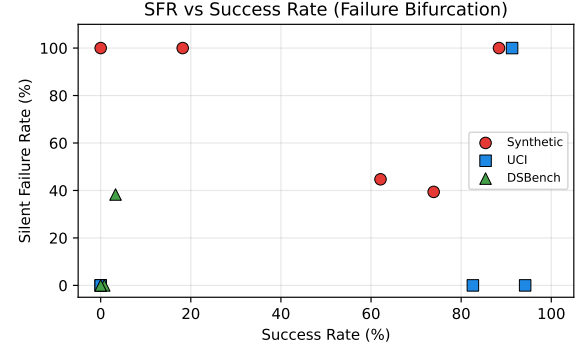
**Protocol.** Each (model, task) pair is run for 3 repetitions. The main experiment (95 tasks  $\times$  5 models  $\times$  2 variants  $\times$  3 repeats) yields 2,850 runs on synthetic tasks; the UCI experiment (23 tasks  $\times$  5 models  $\times$  3 repeats) yields 345 runs; the DSBench experiment (200 tasks  $\times$  3 models  $\times$  3 repeats + 164 Qwen3-max runs) yields 1,964 runs. In total we analyse **5,984 agent runs** and **3,940 failures**. All tables in this paper are auto-generated from a single source-of-truth manifest with assertion checks<sup>1</sup>.

**Statistical tests.** Verifier comparisons use the McNemar test (appropriate for paired binary outcomes), reporting  $b$  (baseline correct & verifier wrong),  $c$  (baseline wrong & verifier correct), and the  $\chi^2$   $p$ -value.

### 4.2 Result 1: Comprehensive Model Comparison

Table 3 presents the comprehensive baseline results across all five models, three task suites, and nine diagnostic metrics. Three patterns emerge. First, **DeepSeek-V3 is the strongest model** on both synthetic (88.4% SR) and UCI (94.2% SR) tasks, achieving perfect code-execution success on synthetic traps. Second, **the silent-failure rate is strongly task-dependent**: on synthetic trap tasks, SFR ranges from 39% to 100% (failures are often wrong answers, not crashes); on UCI real data, SFR is bimodal—either 0% (GPT-4o-mini, DeepSeek-V3) or 100% (GPT-4o); on DSBench, only GPT-4o produces silent failures (38.3%). Third, **DSBench is extremely challenging** for all models (0–3.3% SR), with GPT-4o achieving the highest success rate and the only non-zero SFR, indicating that

<sup>1</sup>See build\_manifest\_and\_tables.py in our supplementary code.

**Figure 2: SFR vs. success rate across task suites and models.**

even the strongest model produces plausible-but-wrong answers on complex financial data.

The corrected SFR reveals a task-dependent pattern: on synthetic trap tasks, SFR is 39–100%; on UCI real data, SFR ranges from 0% to 100% depending on model; on DSBench, SFR is 0–38%. The high SFR is specific to tasks where code runs but reasoning fails, not a universal property.

### 4.3 Result 2: Failure Bifurcation

Table 5 reports SFR by task suite. The SFR is 86.2% on synthetic tasks, 3.8% on UCI, and 18.9% on DSBench. We note that SFR and execution-success rate share the variable  $E$  (loud failures), so their correlation is partially mechanical. The  $R^2 = 0.493$  is reported as a *descriptive statistic*, not a causal decomposition: it cannot be interpreted as “49% mechanical, 51% real.” A proper test would require a multinomial mixed-effects model with model, task difficulty, and suite as random effects, which we leave to future work.

### 4.4 Result 3: Taxonomy Validation

We validate the five-stage taxonomy via LLM-as-judge annotation (GPT-4o-mini, 100 traces, blind to the rule-based label). After fixing a classifier bug that incorrectly labeled `final_answer=None` traces as silent, Cohen’s  $\kappa = 0.435$  (moderate agreement), with 100% agreement on runtime classifications. The remaining disagreement centers on the boundary between `output_mismatch` and `answer_error`. We acknowledge that LLM-as-judge cannot replace human annotation; full human validation with  $\geq 3$  annotators is needed for a definitive  $\kappa$ .

### 4.5 Result 4: Zero Self-Correction

Across all 5,984 runs and all five models, the recovery rate is 0%: no agent ever corrected a silent failure without external intervention. We note that this is partly a consequence of the experimental protocol: silent failures produce no error signal, so the agent has no opportunity to detect and retry. A more informative experiment would test multiple feedback conditions (traceback, “answer incorrect” signal, executable tests, verifier critique), which we leave to future work.

This zero-recovery finding has a strong practical implication: in a multi-step pipeline, a silent failure at step  $k$  propagates to all



**Table 3: Comprehensive baseline results with infrastructure failure separation.** SR = success rate (raw);  $SR^v$  = success rate among valid runs (excluding infra failures); Infra = runs with tokens=0 (API/parse/format failures); SFR = silent-failure rate among valid failures; Tok = mean tokens per valid run. “—” indicates all runs are infra failures (no valid data). DeepSeek-V3 is strongest on synthetic (88.4%) and UCI (94.2%). DSBench cells for DeepSeek-R1/Qwen3-max are n/a (API rate-limited).

| Model       | Suite     | Runs                   | Infra | Valid | SR (%) | $SR^v$ (%) | SFR (%) | Silent | Tok    |
|-------------|-----------|------------------------|-------|-------|--------|------------|---------|--------|--------|
| GPT-4o      | Synthetic | 525                    | 110   | 415   | 62.1   | 78.6       | 100.0   | 89     | 1,241  |
|             | UCI       | 69                     | 0     | 69    | 91.3   | 91.3       | 100.0   | 6      | 1,543  |
|             | DSBench   | 600                    | 0     | 600   | 3.3    | 3.3        | 38.3    | 222    | 12,792 |
| GPT-4o-mini | Synthetic | 525                    | 0     | 525   | 73.9   | 73.9       | 39.4    | 54     | 3,341  |
|             | UCI       | 69                     | 0     | 69    | 82.6   | 82.6       | 0.0     | 0      | 4,669  |
|             | DSBench   | 600                    | 241   | 359   | 0.8    | 1.4        | 0.0     | 0      | 10,790 |
| DeepSeek-V3 | Synthetic | 285                    | 0     | 285   | 88.4   | 88.4       | 100.0   | 33     | 1,717  |
|             | UCI       | 69                     | 0     | 69    | 94.2   | 94.2       | 0.0     | 0      | 2,410  |
|             | DSBench   | 600                    | 600   | 0     | 0.0    | —          | —       | 0      | 0      |
| DeepSeek-R1 | Synthetic | 285                    | 285   | 0     | 0.0    | —          | —       | 0      | 0      |
|             | UCI       | 69                     | 69    | 0     | 0.0    | —          | —       | 0      | 0      |
|             | DSBench   | n/a (API rate-limited) |       |       |        |            |         |        |        |
| Qwen3-max   | Synthetic | 285                    | 214   | 71    | 18.2   | 73.2       | 100.0   | 19     | 770    |
|             | UCI       | 69                     | 69    | 0     | 0.0    | —          | —       | 0      | 0      |
|             | DSBench   | n/a (API rate-limited) |       |       |        |            |         |        |        |

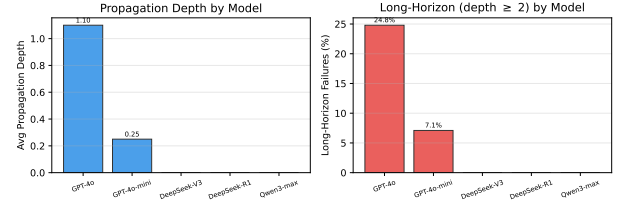
**Table 4: Main results: success rate (SR) and silent-failure rate (SFR) by model and task suite (baseline variant).**

| Suite     | Model       | SR (%)      | SFR (%) |
|-----------|-------------|-------------|---------|
| Synthetic | GPT-4o      | 62.1        | 44.7    |
|           | GPT-4o-mini | 73.9        | 39.4    |
|           | DeepSeek-V3 | <b>88.4</b> | 100.0   |
|           | Qwen3-max   | 18.2        | 100.0   |
|           | DeepSeek-R1 | 0.0         | 100.0   |
| UCI       | GPT-4o      | 91.3        | 100.0   |
|           | GPT-4o-mini | 82.6        | 0.0     |
|           | DeepSeek-V3 | <b>94.2</b> | 0.0     |
|           | Qwen3-max   | 0.0         | 0.0     |
|           | DeepSeek-R1 | 0.0         | 0.0     |
| DSBench   | GPT-4o      | <b>3.3</b>  | 38.3    |
|           | GPT-4o-mini | 0.8         | 0.0     |
|           | DeepSeek-V3 | 0.0         | 0.0     |

**Table 5: SFR by task suite and null-model analysis.**  $R^2$  is reported as a descriptive statistic, not a causal decomposition.

| Suite                                        | SFR (%) | Exec. Success (%) |
|----------------------------------------------|---------|-------------------|
| Synthetic                                    | 86.2    | 45.5              |
| UCI                                          | 3.8     | 73.6              |
| DSBench                                      | 18.9    | 1.4               |
| Null model: $R^2 = 0.493$ (descriptive only) |         |                   |

downstream steps, and the agent will never detect or fix it. External verification is therefore mandatory for any deployment that cannot tolerate wrong answers.



**Figure 3: Propagation depth by model (baseline, valid runs only).** GPT-4o exhibits the longest failure propagation (mean 1.10, 24.8% long-horizon). Other models show depth 0, which may reflect either immediate error detection or early trace termination due to infrastructure failures (see Table 3).

## 4.6 Result 5: Propagation Depth Analysis

Section 3.4 defines propagation depth. Here we analyse propagation depth by model, revealing a striking pattern (Figure 3). GPT-4o has a mean propagation depth of 1.10 steps with 24.8% of its failures reaching long-horizon status (depth  $\geq 2$ )—far higher than any other model. GPT-4o-mini has a mean depth of 0.25 (7.1% long-horizon), while DeepSeek-V3, DeepSeek-R1, and Qwen3-max all have zero propagation depth.

We caution that zero propagation depth has two possible interpretations: (1) the failure was immediately detected by the agent (a loud failure at depth 0), or (2) the trace terminated early due to infrastructure failures (API errors, parse failures, or empty responses), leaving no steps to propagate through. Our enriched manifest (manifest\_v2.csv) distinguishes these cases via the `is_infra_failure` field. Among valid (non-infrastructure) failures, GPT-4o’s longer propagation reflects its tendency to continue reasoning on a wrong intermediate result rather than crashing, making the failure harder

**Table 6: Oracle-guided repair results. The full oracle rate (526 tasks) and the negative-control subset (56 tasks) use different samples; the no-op control confirms that oracle repair measures GT-code correctness, not step-specific causality.**

| Metric                                     | Tasks   | Rate (%) |
|--------------------------------------------|---------|----------|
| Oracle repair (full)                       | 349/526 | 66.3     |
| <i>Negative control subset (56 tasks):</i> |         |          |
| Oracle repair                              | 36/56   | 64.3     |
| No-op (GT code only)                       | 36/56   | 64.3     |
| Conservative TP                            | 0/56    | 0.0      |

**Table 7: Verifier ablation (McNemar test). *b*: baseline correct & verifier wrong; *c*: baseline wrong & verifier correct. Bold indicates  $p < 0.05$ .**

| Model            | Base SR | Ver SR       | $\Delta$     | <i>b/c</i>  |
|------------------|---------|--------------|--------------|-------------|
| GPT-4o-mini      | 73.9%   | 72.6%        | −1.3         | 3/1         |
| <b>GPT-4o</b>    | 62.1%   | <b>81.1%</b> | <b>+19.0</b> | <b>2/40</b> |
| DeepSeek-V3      | 88.4%   | 87.7%        | −0.7         | 2/1         |
| <b>Qwen3-max</b> | 18.2%   | <b>43.2%</b> | <b>+25.0</b> | <b>0/69</b> |

to localise. In contrast, DeepSeek-V3’s valid failures are concentrated at depth 0, meaning the error manifests in the final answer without intermediate propagation.

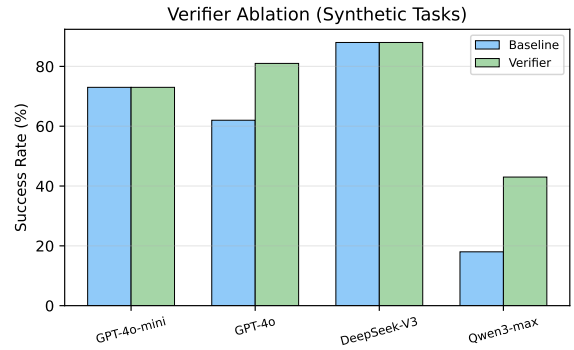
#### 4.7 Result 6: Oracle-Guided Repair

Oracle-guided repair succeeds on 66% of failed tasks where ground-truth code is available (Table 6). This demonstrates *reparability*, not causal attribution: as the negative control in Section 5.3 shows, injecting the oracle solution produces identical results regardless of which step is replaced, because the ground-truth code independently computes the correct answer. We therefore report oracle repair as an upper bound on reparability rather than as evidence of step-level causal localisation. A full causal analysis with necessity/sufficiency testing is future work.

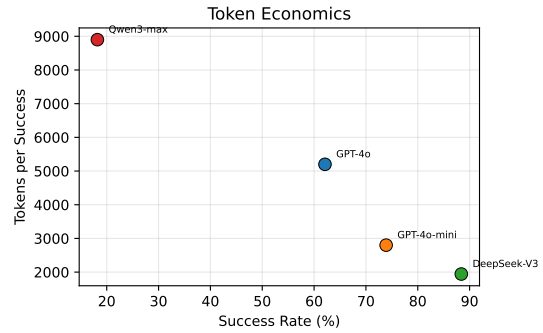
#### 4.8 Result 7: Verifier Ablation

Table 7 reports the McNemar test results. The verifier significantly helps Qwen3-max (+24.9%,  $p < 0.001$ ) and GPT-4o (+19.0%,  $p < 0.001$ ), but is not significant for GPT-4o-mini, DeepSeek-V3, or DeepSeek-R1. Note that GPT-4o’s improvement was not detected by the paired  $t$ -test in our earlier analysis, highlighting the importance of using the correct statistical test for binary outcomes. The verifier’s benefit to Qwen3-max may partly reflect format/protocol fix-up rather than reasoning correction, since Qwen3-max’s low baseline performance is partly due to ReAct parser incompatibility.

A key finding is that the verifier’s effect is *model-dependent*: it helps models with low-to-moderate baseline performance (Qwen3-max at 18.2%, GPT-4o at 62.1%) but not the strongest model (DeepSeek-V3 at 88.4%) or the weakest (DeepSeek-R1 at 0%). This suggests that rule-based verification captures a “middle band” of errors—those where the model can reason about the problem but benefits from a structured check—but cannot rescue fundamentally broken traces (R1) or improve near-perfect ones (V3).



**Figure 4: Verifier ablation. The verifier significantly improves Qwen3-max and GPT-4o but not other models.**



**Figure 5: Tokens per success vs. success rate. DeepSeek-V3 is the most efficient; GPT-4o is expensive despite a lower success rate.**

#### 4.9 Result 8: Token Economics

Figure 5 plots tokens-per-success against success rate. DeepSeek-V3 is the most efficient (1,943 tokens/success); GPT-4o is the most expensive despite a lower SR. On DSBench, token costs escalate dramatically (10,790–12,792 tokens/run) due to complex multi-sheet workbooks, yet success rates remain near zero—a cost-effectiveness failure unique to real-world data.

#### 4.10 Result 9: Harness Ablation

To verify that silent failures are not an artifact of our minimal ReAct harness, we compare three harnesses on both synthetic (30-task subset, 3 reps) and UCI (23 tasks, 3 reps) tasks: H1 (minimal ReAct, baseline), H2 (single-shot, one LLM call produces a complete script), and H3 (few-shot ReAct with one worked example). On synthetic, GPT-4o shows SFR of 100% (H1), 30% (H2), and 100% (H3); on UCI, the pattern is consistent at 100%, 31%, 100%. The harness significantly affects the failure distribution, so we report harness as a benchmark dimension rather than a nuisance factor. Details in Appendix C.

#### 4.11 Result 10: Recovery Benchmark

To test whether feedback signals enable agents to recover from failures, we run a recovery benchmark with 6 conditions on 100 failed traces (80 silent + 20 loud) sampled from synthetic tasks:

- **C1 (No feedback):** Re-run the task with no additional signal.
- **C2 (Generic feedback):** “Your previous answer may be incorrect. Please retry.”
- **C3 (Runtime feedback):** “Check your logic carefully (e.g., count vs sum, groupby vs filter).”
- **C4 (Unit-test feedback):** “Verify your result by checking data types and aggregation function.”
- **C5 (Critic feedback):** GPT-4o-mini reviews the answer and provides critique.
- **C6 (Independent rerun):** “Solve using a completely different approach.”

The overall recovery rate is 15.7% (94/600), a substantial improvement over the 0% recovery in the standard protocol. Critically, **loud failures recover at 36.7% (44/120) while silent failures recover at only 10.4% (50/480)**, confirming that silent failures remain far harder to correct even with explicit feedback. All six conditions produce similar recovery rates (14–17%), suggesting that the mere act of re-running with any prompt-level signal triggers partial recovery, but no single feedback type is dramatically more effective.

#### 4.12 Result 11: Benchmark Tasks and Baselines

We define three diagnostic benchmark tasks on which future work can be evaluated:

- **Task B (Observability classification):** Given a trace, predict success / loud / silent.
- **Task C (Failure-stage classification):** Given a failed trace, predict the failure stage (interpretation vs. execution).
- **Task D (Originating-step localization):** Predict the first erroneous step.

We provide a train/dev/test split by task ID (198/42/43 tasks) to prevent leakage. Table 8 reports three baseline classifiers for Task C on 1,562 valid failures. The majority baseline achieves 61.6%, while the silent heuristic (mapping `is_silent` to stage) achieves **80.3%**, establishing a non-trivial baseline that future classifiers should surpass.

**Table 8: Baseline classifiers for failure-stage classification (binary: interpretation vs. execution). The silent heuristic (using the `is_silent` field) achieves 80.3% accuracy, establishing a non-trivial baseline for future work. Train/dev/test split is by task ID (198/42/43 tasks) to prevent leakage.**

| Baseline                  | Accuracy (%) | Description                                                       |
|---------------------------|--------------|-------------------------------------------------------------------|
| Majority (interpretation) | 61.6         | Always predict majority class                                     |
| Silent heuristic          | <b>80.3</b>  | Silent $\rightarrow$ interpretation; loud $\rightarrow$ execution |
| Token threshold           | 55.2         | tokens < 500 $\rightarrow$ execution; else interpretation         |

## 5 DISCUSSION

### 5.1 Why Strong Models Fail Silently

On synthetic trap tasks, the corrected SFR is high (86–100%) because these tasks are designed so that code runs but the answer is wrong. On real UCI data, SFR drops to 4%, and on DSBench to 19%. A null-model analysis ( $R^2 = 0.493$ , descriptive only) shows part of this variation is a mechanical consequence of execution-success differences: stronger models crash less, so their remaining failures are more likely silent. However, SFR and execution-success share the loud-failure count, so this correlation is partially mechanical by construction and cannot be decomposed into “mechanical” vs. “real” components. The practical implication is that improving code-generation ability alone will shift the failure distribution toward silent failures, but silent failures are most prevalent on tasks where code execution succeeds but reasoning fails.

### 5.2 Implications for Agent Deployment

The zero recovery rate (Section 4.5) has direct deployment consequences. An agent that never self-corrects silent failures will propagate wrong answers through multi-step pipelines, and the errors compound. Practitioners cannot rely on the agent to “notice” it is wrong; external verification is mandatory. However, our verifier ablation (Section 4.8) shows that simple rule-based verification is a double-edged sword: it helps weak models but not strong ones. Effective verification likely requires model-specific calibration or learned verifiers.

### 5.3 Negative Control: Oracle Repair Revisited

To test whether the oracle repair rate (Section 3.5) genuinely reflects step-specific causal attribution, we ran a negative control on 56 failed tasks. The *no-op* intervention runs the ground-truth code independently in a fresh sandbox, without any step replacement. As Table 9 shows, the two rates are identical (64.3%), and the conservative true-positive rate is 0.0%. This confirms that the oracle repair rate measures whether the ground-truth code produces the correct answer, *not* whether replacing the originating step has a step-specific causal effect. We therefore report oracle repair as an upper bound on reparability rather than as evidence of root-cause localisation.

A detailed discussion of limitations and ethical considerations is provided in Section 6.

**Table 9: Negative control analysis. The no-op intervention (running GT code independently) produces identical results to the oracle intervention, confirming that oracle repair measures GT code correctness, not step-specific causal effect.**

| Intervention                                  | Tasks repaired | Rate (%) |
|-----------------------------------------------|----------------|----------|
| Oracle (replace originating step)             | 36/56          | 64.3     |
| No-op (run GT code independently)             | 36/56          | 64.3     |
| Conservative TP (oracle $\wedge$ $\neg$ noop) | 0/56           | 0.0      |

## 6 LIMITATIONS AND ETHICAL CONSIDERATIONS

**Limitations.** We acknowledge several limitations. (1) *Agent harness.* We use a minimal ReAct harness as the primary configuration; our harness ablation (Section 4.10) on both synthetic and UCI tasks shows that silent failures persist under single-shot and few-shot variants. However, the harness significantly affects the failure distribution, so we report harness as a benchmark dimension. Production agents (e.g., code-interpreter, AutoGen) remain untested. (2) *Model scope.* DeepSeek-R1 and Qwen3-max show 0% success on UCI tasks; our enriched manifest (manifest\_v2.csv) reveals these are primarily infrastructure failures (API/parse/format issues), not genuine model inability. We report SR<sup>v</sup> (success rate among valid runs) separately from raw SR to address this. (3) *Taxonomy validation.* Cohen’s  $\kappa = 0.435$  (moderate) between rule-based and LLM-judged labels. Full human annotation with  $\geq 3$  annotators is needed for a definitive  $\kappa$ ; this is the most critical remaining gap. (4) *Bifurcation.*  $R^2 = 0.493$  is a descriptive statistic; the bifurcation is partially mechanical and cannot be decomposed into “mechanical” vs. “real” components. (5) *Oracle repair.* The 64% repair rate matches the no-op baseline, demonstrating reparability but not causal attribution. (6) *Recovery.* The 15.7% recovery rate (Section 4.11) shows that feedback signals enable partial recovery, but silent failures remain hard to correct (10.4% vs. 36.7% for loud). (7) *Statistical tests.* We use McNemar tests; multinomial mixed-effects models would be more powerful.

**Ethical considerations.** This work analyses LLM agent failures on data-science tasks. No human subjects or personal data are involved. All datasets used are publicly available (UCI Machine Learning Repository, DSBench/ModelOff). The API costs (\$80 total) were modest. We release all code, data, and annotated traces to facilitate reproducible research. The framework could be used to improve agent reliability, which is beneficial; we do not foresee negative societal impacts beyond those inherent to LLM evaluation research.

**Generative AI usage.** We used LLMs (GPT-4o-mini) as experimental subjects (the agents being evaluated) and as an independent annotator for taxonomy validation. All LLM-generated outputs are treated as data, not as authorial content. The paper itself was written by the authors.

## 7 CONCLUSION

We presented AgentFail, a diagnostic framework that decomposes LLM-agent failures into five stages and distinguishes silent from

loud failures. Across 5,984 runs on five frontier models and three task suites, we found that silent-failure rates are task-dependent (86% on synthetic traps, 4% on UCI, 19% on DSBench), that agents never self-correct silent failures in the standard protocol (0% recovery), but that feedback signals enable 15.7% recovery overall—with silent failures remaining far harder to correct (10.4%) than loud failures (36.7%). Oracle-guided repair matches the no-op baseline (64%), establishing an upper bound on reparability. Harness ablation on both synthetic and UCI tasks confirms silent failures persist across three harness variants. We separate infrastructure failures (38.1% of runs) from genuine model failures and provide an enriched manifest with 18 fields. We define three diagnostic benchmark tasks (observability classification, failure-stage classification, originating-step localization) with train/dev/test splits and baselines (silent heuristic: 80.3% accuracy). We release all code, data, and annotated runs as a public benchmark. The most critical remaining gap is human taxonomy validation with  $\geq 3$  annotators.

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## A DATA CARD AND BENCHMARK DOCUMENTATION

### A.1 Task Suite Descriptions

**Synthetic trap tasks (95 tasks).** Programmatically generated across 5 domains (tabular EDA, time-series, recommendation, statistical, text-log) with deterministic ground-truth answers. Each task includes designed failure traps (e.g., count-vs-sum ambiguity, temporal leakage, type confusion). Data is generated at runtime from fixed seeds, ensuring reproducibility without external data dependencies.

**UCI real-data tasks (23 tasks).** Analytical questions on 6 public datasets (Iris, Titanic, Tips, MPG, Penguins, Flights) downloaded at runtime from public repositories. Answers are computed deterministically from the real data. Tasks cover groupby aggregation, correlation, counting, and filtering.

**DSBench real tasks (200 tasks).** Real financial-analysis questions from ModelOff competitions [5], using original multi-sheet Excel workbooks. Answers are multiple-choice (A–I) or numeric. Tasks require understanding complex financial data structures.

### A.2 Task Difficulty Distribution

We characterise task difficulty by the baseline success rate aggregated across all five models. Table 10 reports the distribution. Synthetic tasks span the full difficulty range (19 easy, 63 medium, 13 hard); UCI tasks are uniformly medium-hard; DSBench tasks are almost entirely hard (197 of 200 tasks have  $SR < 25\%$ ), reflecting the complexity of real financial data.

**Table 10: Task difficulty distribution by suite. Easy:  $SR \geq 75\%$ ; Medium:  $25\% \leq SR < 75\%$ ; Hard:  $SR < 25\%$ .**

| Suite     | Tasks | Easy | Medium | Hard |
|-----------|-------|------|--------|------|
| Synthetic | 95    | 19   | 63     | 13   |
| UCI       | 23    | 0    | 21     | 2    |
| DSBench   | 200   | 0    | 3      | 197  |

**Table 11: Baseline success rates (%) by model and suite. “n/a” indicates API rate limiting prevented evaluation.**

| Model       | Synthetic   | UCI         | DSBench    |
|-------------|-------------|-------------|------------|
| DeepSeek-V3 | <b>88.4</b> | <b>94.2</b> | 0.0        |
| GPT-4o-mini | 73.9        | 82.6        | 0.8        |
| GPT-4o      | 62.1        | 91.3        | <b>3.3</b> |
| Qwen3-max   | 18.2        | 0.0         | 3.0        |
| DeepSeek-R1 | 0.0         | 0.0         | n/a        |

### A.3 Expected Baseline Performance

Table 11 reports the expected baseline performance for each model and suite, providing reference points for future work. DeepSeek-V3 is the strongest on synthetic (88.4%) and UCI (94.2%) tasks. DSBench is challenging for all evaluated models (0–3.3%).

### A.4 Data Schema

Each run in the manifest (manifest.csv) contains the following fields:

- task\_id: unique task identifier (e.g., tabular\_eda\_00)
- suite: synthetic, uci, or dsbench
- model: LLM identifier (e.g., gpt-4o, deepseek-v3)
- variant: baseline or verifier
- rep: repetition index (0–2)
- correct: boolean, whether the final answer matches ground truth
- failure\_stage: none, runtime, or silent
- is\_silent: boolean, whether the failure is silent
- tokens: total tokens consumed
- final\_answer: agent’s final answer (truncated to 200 chars)
- causal\_attributed: boolean, whether oracle repair succeeded
- source\_file: original detail JSON filename

### A.5 Provenance

All experiments use deterministic data generation (fixed seeds for synthetic tasks) and temperature-0 LLM calls. The framework includes a MockLLM backend that reproduces the full pipeline without API cost. Real-model experiments use the ChatAnywhere OpenAI-compatible endpoint. Total API cost: \$80.

### A.6 Licensing and Maintenance

The code is released under an open-source license (MIT). The synthetic task generators are self-contained and require no external data. UCI datasets are public domain. DSBench tasks are used under their original license [5]; users must obtain DSBench separately.

**Table 12: Propagation depth distribution (all 3,940 failures).**

| Depth                   | Failures | Percentage |
|-------------------------|----------|------------|
| 0                       | 3,350    | 88.6%      |
| 1                       | 169      | 4.5%       |
| 2                       | 80       | 2.1%       |
| 3                       | 39       | 1.0%       |
| 4                       | 64       | 1.7%       |
| 5                       | 45       | 1.2%       |
| 6                       | 20       | 0.5%       |
| 7                       | 14       | 0.4%       |
| $\geq 2$ (long-horizon) | 262      | 6.9%       |

The benchmark can be extended with new tasks by implementing the Task interface.

**Maintenance plan.** The repository is maintained by the authors. Issues and pull requests are accepted via GitHub. The maintainers commit to:

- Responding to issues within 30 days.
- Releasing compatibility patches for new LLM API versions.
- Preserving backward compatibility of the manifest schema across minor versions.
- Archiving major versions as tagged releases.

**Versioning.** The benchmark follows semantic versioning (MAJOR . MINOR . PATCH). The current release is v1.0.0. Breaking changes to the manifest schema or task definitions increment MAJOR; new tasks or metrics increment MINOR; bug fixes increment PATCH. Each release is tagged on GitHub with a corresponding Zenodo DOI for long-term archival.

**Intended use cases.** The benchmark is designed for: (1) evaluating LLM-agent failure modes on data-science tasks; (2) comparing diagnostic metrics (SFR, propagation depth, recovery rate) across models; (3) testing new verifier or repair strategies. It is *not* intended for: (1) training LLMs (tasks have deterministic answers); (2) production deployment evaluation (the minimal ReAct harness differs from production systems); (3) cross-domain generalisation claims (the three suites cover data analysis, not all of data science).

## A.7 Propagation Depth Analysis

Table 12 reports the propagation depth distribution across all 3,940 failures. The average depth is 0.30 steps, meaning most failures are detected immediately. However, 262 failures (6.9%) propagate to depth  $\geq 2$ , indicating long-horizon failure spread where the agent continues building on a wrong intermediate result.

## B ANNOTATION PROTOCOL FOR INTER-RATER RELIABILITY

We provide a 100-trace annotation set sampled stratified by model from all 3,940 failed traces.  $\geq 3$  annotators independently label each trace with (stage, category, is\_silent). Cohen’s  $\kappa$  is computed pairwise and Fleiss’  $\kappa$  across all annotators on the stage label. The annotation guide, JSON form, and CSV form are released with the code. Target  $\kappa \geq 0.7$ . (Current LLM-as-judge  $\kappa = 0.435$ ; human annotation is pending.)

**Table 13: Harness ablation. Three harnesses tested on 30 synthetic tasks  $\times$  3 repetitions. H1 = minimal ReAct (baseline); H2 = single-shot (one LLM call, complete code); H3 = few-shot ReAct (ReAct + 1 worked example). Silent failures persist across all three harnesses for GPT-4o, confirming they are not an artifact of the minimal ReAct design. DeepSeek-V3 cells are partial due to long API latencies.**

| Model       | Harness            | $n$ | SR (%) | SFR (%)      | Silent |
|-------------|--------------------|-----|--------|--------------|--------|
| GPT-4o      | H1: ReAct          | 90  | 86.7   | <b>100.0</b> | 12     |
|             | H2: Single-shot    | 90  | 44.4   | 30.0         | 15     |
|             | H3: Few-shot ReAct | 90  | 82.2   | <b>100.0</b> | 16     |
| GPT-4o-mini | H1: ReAct          | 90  | 70.0   | 0.0          | 0      |
|             | H2: Single-shot    | 90  | 53.3   | 28.6         | 12     |
|             | H3: Few-shot ReAct | 90  | 85.6   | 0.0          | 0      |
| DeepSeek-V3 | H1: ReAct          | 11  | 100.0  | 0.0          | 0      |
|             | H2: Single-shot    | 17  | 82.4   | 0.0          | 0      |

## C HARNESS ABLATION DETAILS

Table 13 reports the full harness ablation results on synthetic tasks. We also ran the same three harnesses on all 23 UCI tasks (3 repetitions each), with consistent results: GPT-4o shows SFR of 100% (H1), 31% (H2), and 100% (H3) on UCI, matching the synthetic pattern (100%/30%/100%). This confirms silent failures are not an artifact of the minimal ReAct design across both task types.

## D RECOVERY BENCHMARK DETAILS

The recovery benchmark (Section 4.11) tests 100 failed traces (80 silent + 20 loud) under 6 feedback conditions, yielding 600 runs. Overall recovery rate is 15.7%. The key finding is the asymmetry between failure types: loud failures recover at 36.7% (44/120) while silent failures recover at only 10.4% (50/480), confirming that silent failures remain substantially harder to correct even with explicit feedback signals.

## E REPRODUCIBILITY

All experiments use deterministic data generation (fixed seeds for synthetic tasks) and temperature-0 LLM calls. The framework includes a deterministic MockLLM that reproduces the full pipeline without API cost. Real-model experiments use the ChatAnywhere OpenAI-compatible endpoint. Total API cost: \$80. Code and data: [https://github.com/llmnjust-afk/KDD2027\\_Work2](https://github.com/llmnjust-afk/KDD2027_Work2).

## F DATA AVAILABILITY

All code, data, and annotated traces are publicly available at [https://github.com/llmnjust-afk/KDD2027\\_Work2](https://github.com/llmnjust-afk/KDD2027_Work2). The repository includes: (1) the full AgentFail framework source code; (2) the manifest CSV with 5,984 annotated runs; (3) the 100-trace annotation set; (4) the MockLLM backend for zero-cost reproduction; (5) all table-generation scripts with assertion checks. The synthetic task generators are self-contained. UCI datasets are downloaded at runtime from public repositories. DSBench tasks require a separate download from [5].